

MOHANRAJ E

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Education

Mailam Engineering College (Anna University)

Sep 2018 – July 2022

Bachelor of Engineering in Computer Science and Engineering (B.E – CSE) | CGPA: 7.64 | Villupuram, Tamil Nadu

Relevant Coursework: Python Programming | DOTNET Technology | Web Designing & Web Application Development | Android Studio | Node.js | Express.js | MongoDB | GitLab CI/CD | Qt 5.5 | Tesseract OCR

Experience

Software Engineer with 2+ years building high-concurrency Qt/C++ desktop applications and OCR pipelines. Proficient in C++17, Python, full-stack web, and relational databases. Applies SOLID principles and multi-threading to deliver maintainable, production-grade systems.

Quantix Technology & Consulting Services, LLP

Feb 2024 – Mar 2024

Front-End Developer | Construction Site Website | React, Tailwind CSS, Vite

- Developed a responsive construction site website using React and Vite, implementing component-based architecture for scalable and maintainable UI.
- Styled the application using Tailwind CSS, ensuring a consistent, mobile-first design with fast rendering and minimal CSS overhead.

Wipro, Kochi

Oct 2024 – Jul 2025

Project Engineer | Printer Management System | Qt 5.5, C++17, SQLite, TCP/IP Sockets, Mutex Threads, Bcrypt, MVVM Architecture

- Engineered an end-to-end printer scheduling application using Qt 5.5 and C++17, successfully serving 50+ simultaneous client nodes with zero data conflicts.
- Architected a 5-layer modular N-tier architecture (UI, ViewModel, Command, Manager, DAO) applying MVVM, Factory, and Facade patterns to isolate full-stack responsibilities.
- Built a multi-threaded TCP/IP network layer with mutex-synchronized handlers, establishing real-time backend-to-client status broadcasts.
- Optimized an ACID-compliant SQLite data layer utilizing transaction-based conflict resolution, completely eliminating hardware double-booking errors.
- Implemented secure authentication pipelines using bcrypt password hashing and Role-Based Access Control (RBAC) to protect permission-gated administrative operations.

Wipro, Kochi

Aug 2025 – Nov 2025

Project Engineer | Intelligent OCR Document Processing Suite | Qt, C++, Tesseract OCR Engine, PDF Parsing, Qt Print Support, UTF-8 Encoding

- Developed high-performance OCR suite integrating Tesseract engine into Qt/C++ framework, achieving full text extraction coverage across scanned and digital documents.
- Engineered hybrid extraction logic with automatic fallback from direct PDF stream parsing to Tesseract OCR, delivering 100% document-type compatibility.
- Designed pluggable saver interface supporting UTF-8 and PDF output via Qt Print Support, standardizing document generation across all export formats.

Wipro, Kochi

Jan 2026 – Apr 2026

Project Engineer | FoIP (Fax over IP) Protocol Emulator | C++, Qt, QML, T.30 Protocol, Ghostscript, TCP/IPC Sockets, LibXLSXwriter

- Designed and developed a FoIP emulator using C++ and Qt/QML, implementing a modular, event-driven architecture for real-time fax communication with T.30 protocol simulation, custom state machine, and capability negotiation (DIS/DCS) for optimized transmission parameters.
- Engineered a media pipeline for PDF/TIFF processing using Ghostscript and built a frame-based TCP/IPC communication layer using Qt sockets for reliable data transport.
- Developed session orchestration, logging, retry mechanisms, and XLSX→PDF conversion integration, ensuring end-to-end transmission reliability and full traceability.

Technical Skills

Languages: C++17, C#, Python, JavaScript

Desktop: Qt 5.5, QML, Qt Creator, MVVM, Multithreading, Design Patterns

Web: HTML, CSS, JavaScript, DOM, Angular, React, Tailwind CSS, Vite

Backend: Node.js, Express.js, REST API, CRUD, TCP/IP, JSON, Client-Server Architecture

Databases: SQLite, SQL, MySQL, MongoDB, ACID Transactions

Platforms: .NET, Android Studio

Tools: Git, GitLab CI/CD, bcrypt, Tesseract OCR, Qt Concurrent, Qt Print Support, Ghostscript

Certifications & Training

- Python Programming — Accord Info Matrix Pvt Ltd, Chennai | Cert. No: AIM PY 16968 | Nov 2023
- DOTNET Technology (C#.NET, ADO.NET, ASP.NET, MVC) — Accord Info Matrix Pvt Ltd, Chennai | Cert. No: AIM DT 16969 | Nov 2023
- Web Designing & Web Application Development (HTML, CSS, JavaScript, Angular, DOM) — Accord Info Matrix Pvt Ltd, Chennai | Cert. No: AIM WD 16970 | Nov 2023
- Wipro RPS Training (3 months) — Application Development, Android Studio, Node.js, Express.js, MongoDB
- GitLab CI/CD: Pipelines, CI/CD and DevOps for Beginners — Udemy | May 2026

Leadership & Extracurricular

Cricket: Zone-Level Winner | Kabaddi: House-Match Champion | Friends of Police Volunteer | Cancer Awareness Marathon, Puducherry